

Steiner Analysis

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The introduction of the Downs analysis stimulated a number of enthusiastic investigators and clinicians to develop their own analyses. The spate of analyses that followed served little more than to confuse the issue for clinicians; simply too many landmarks were identified and too many measurements were advocated. As a result, meaningful information was submerged in a quagmire of inconsequential details.

Cecil C. Steiner, however, selected what he considered to be the most meaningful parameters and developed a composite analysis, which he believed would provide the maximum clinical information with the fewest number of measurements.

Certain measurements were then selected, and the means or averages were determined on a series of individuals with normal occlusions. By comparing the traced readings or measurements of patients with malocclusions to those of "normal" occlusions, the degree of deviation from the normal could be determined.

Three-Way Analysis

In the assessment of a lateral cephalometric headfilm, Steiner proposed the appraisal of various parts of the skull separately, namely the skeleton, dentition, and soft tissues. The skeletal analysis entails relating the mandible and maxilla to the skull and to each other. The dental analysis involves relating the maxillary and mandibular incisors to their respective jaws and to each other. Finally, the soft tissue analysis provides a means of assessing the balance and harmony of the lower facial profile.¹⁻³

Skeletal analysis

As described in chapter 4, the lateral cephalometric headfilm is traced, and the traditional landmarks and planes are identified (Figs 7-1 and 7-2). The conventional plane used

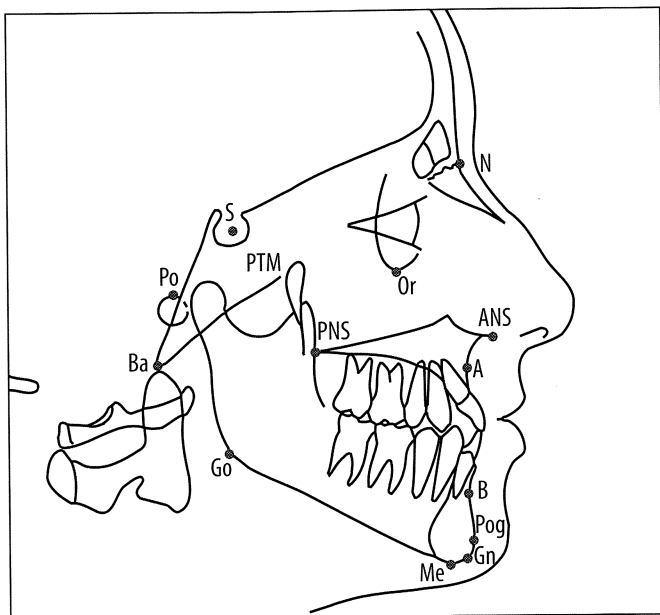


Fig 7-1 Traditional lateral cephalometric headfilm landmarks used with Steiner analysis. Ba = basion; Po = porion; S = sella; PTM = pterygomaxillare; Or = orbitale; N = nasion; PNS = posterior nasal spine; ANS = anterior nasal spine; A = point A; B = point B; Pog = pogonion; Gn = gnathion; Me = menton; Go = gonion.

by anthropologists (and Downs) for relating craniofacial structures when studying dry skulls is the Frankfort horizontal. On lateral cephalometric headfilms, however, landmarks such as porion and orbitale are not always easily identified. Consequently, Steiner elected to use the anterior cranial base (sella to nasion [SN]) as the line of reference to which the jaws would be related. The advantage of using these two midline points is that they are moved only a minimal amount whenever the head deviates from the true profile position. This remains true even if the head is rotated in the cephalostat.

Having established a plane of reference, the anteroposterior apical base relationship of the maxilla and mandible to each other and to the anterior cranial base can now be determined.

Maxilla

Points A and B are regarded as the anterior limits of the apical bases of the maxilla and mandible, respectively. Point A is not an ideal reference point; nevertheless, it is still widely used. (For further discussion and suggestions on identification of this landmark, the reader is referred to Jacobson and Jacobson.⁴) Therefore, to determine whether the maxilla is positioned anteriorly or posteriorly to the cranial base, the angle sella–nasion–point A (SNA) is noted. The mean SNA reading is 82 degrees (Fig 7-3, *a*); thus, if the angular reading is greater than 82 degrees, it would indicate a relative forward position-

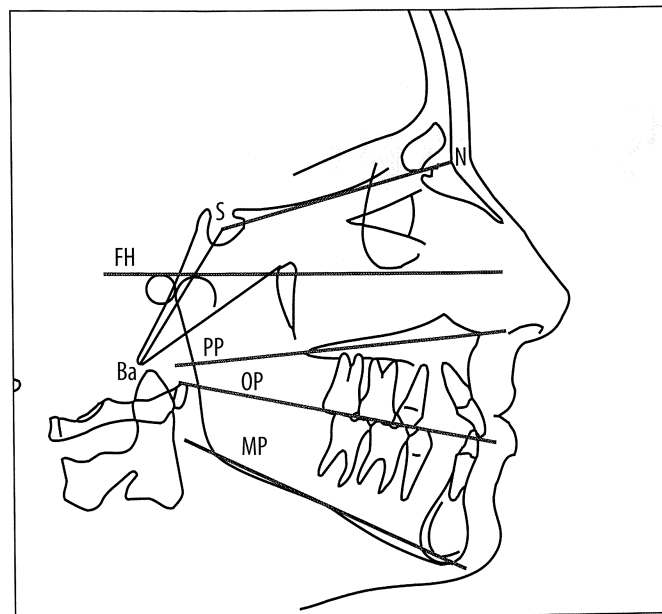


Fig 7-2 Traditional lateral cephalometric headfilm planes used with Steiner analysis. FH = Frankfort horizontal plane; PP = palatal plane; OP = occlusal plane; MP = mandibular plane.

ing of the maxilla (Fig 7-3, *b*). Conversely, if the reading is less than 82 degrees, it would indicate a relative backward or recessive location of the maxilla (Fig 7-3, *c*).

Mandible

To assess whether the mandible is protrusive or recessive relative to the cranial base, the sella–nasion–point B (SNB) angle is read (mean, 80 degrees) (Fig 7-4, *a*). An angle less than 80 degrees indicates a recessive mandible (Fig 7-4, *b*). An angle greater than 80 degrees suggests a prognathic mandible (Fig 7-4, *c*).

Maxilla-mandible relationship

By noting the SNA and SNB readings, the offending jaw can usually be pinpointed. The more significant reading, however, is the point A–nasion–point B (ANB) reading, which provides information on the position of the jaws relative to each other.

Steiner stated that he is "not greatly concerned about the angle SNA because it merely shows whether the face protrudes or retrudes below the skull."¹ Of major concern to Steiner, however, is the difference between SNA and SNB, or the ANB angle. The ANB angle provides a general idea of the anteroposterior discrepancy of the maxillary to the mandibular apical bases. The mean reading for this angle is 2 degrees (Fig 7-5); a reading greater than 2 degrees indicates a Class II skeletal tendency. As a rule, the larger the figure, the greater the antero-

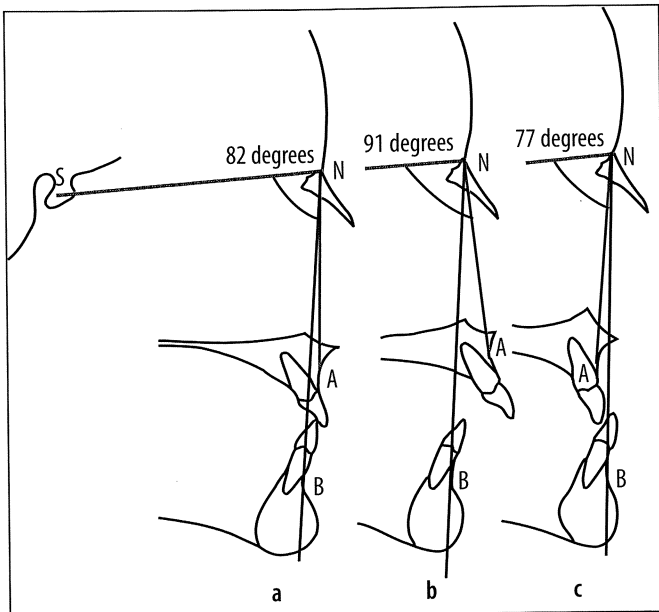


Fig 7-3 SNA angle. (a) The mean SNA reading is 82 degrees. (b) An SNA angle of 91 degrees suggests a protrusive maxilla. (c) An SNA angle of 77 degrees suggests a recessive maxilla.

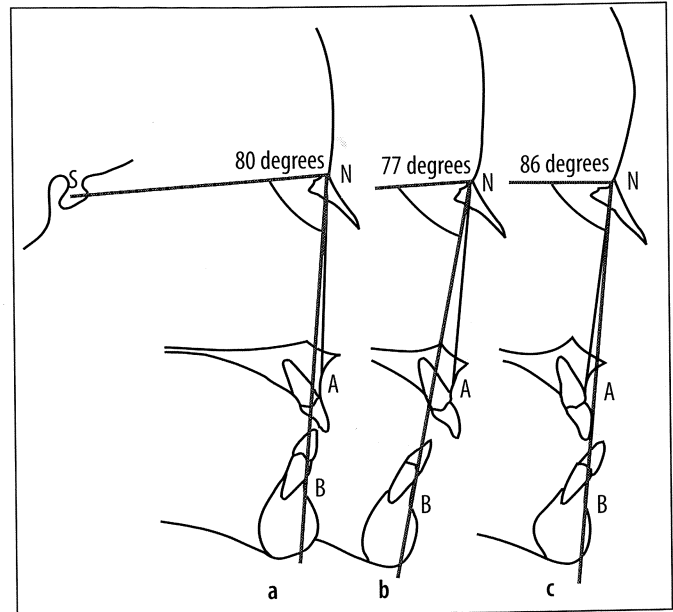


Fig 7-4 SNB angle. (a) The mean SNB reading is 80 degrees. (b) An SNB angle of 77 degrees suggests a recessive mandible. (c) An SNB angle of 86 degrees suggests a protrusive mandible.

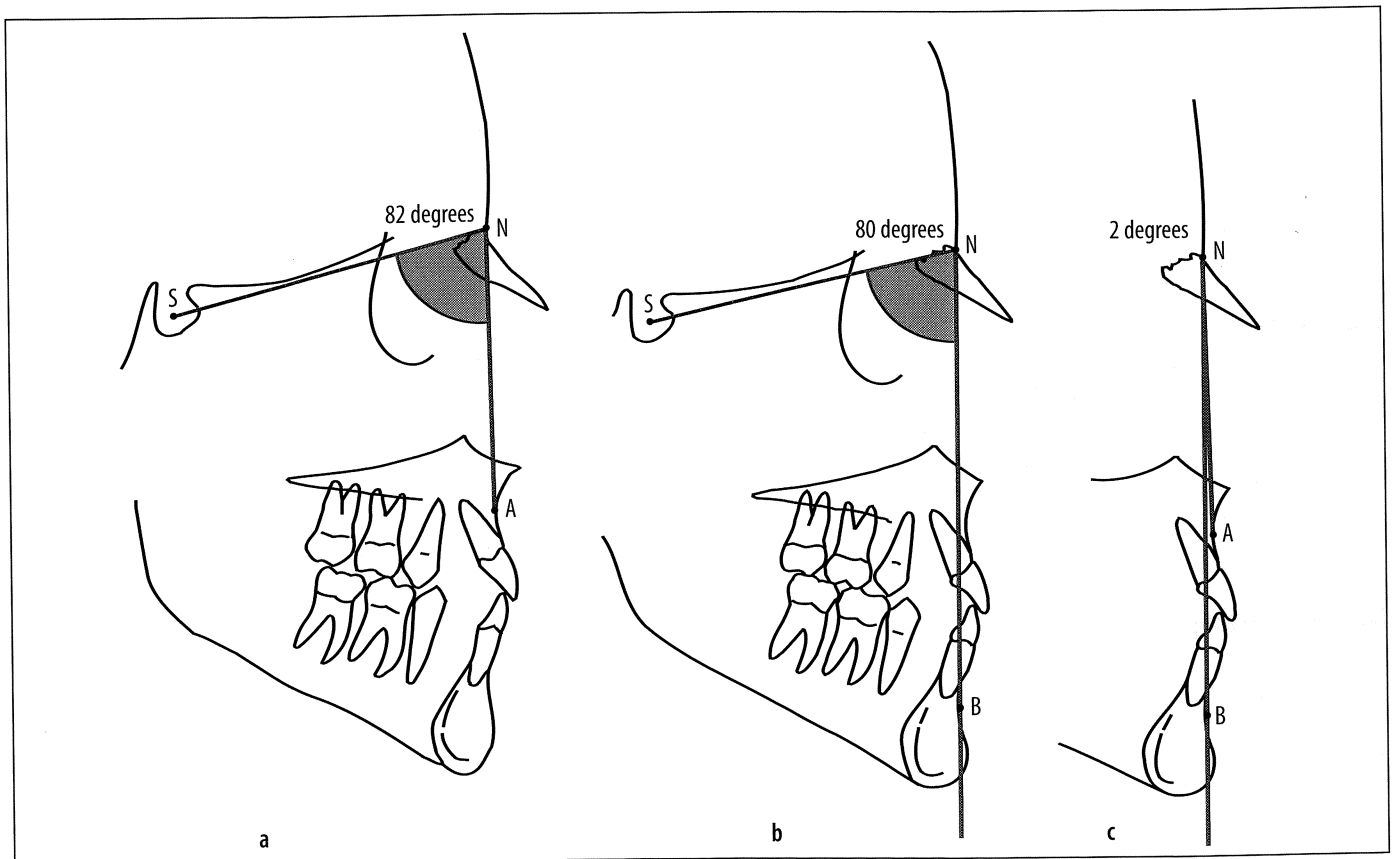


Fig 7-5 ANB angle. The mean ANB angle of 2 degrees (c) is the difference between the SNA (a) and SNB (b) angles in "normal" occlusions.

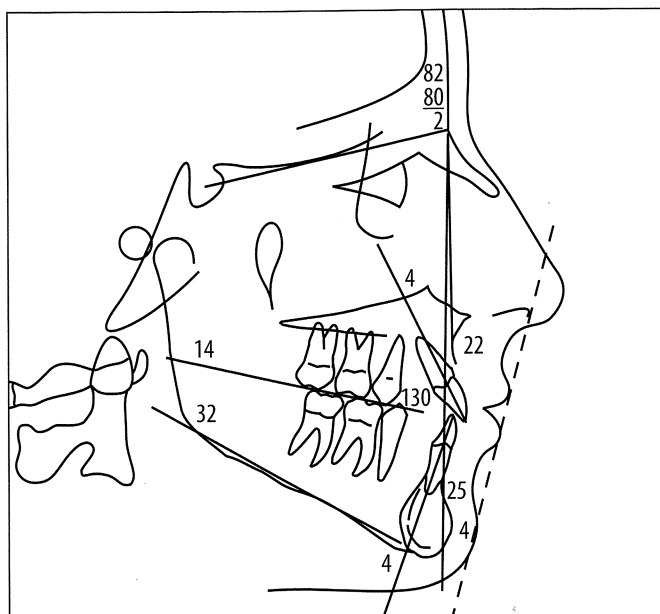


Fig 7-6 Various planes and angles of the Steiner analysis drawn and measured. By tradition, the measurement figures are located on the tracing, as in this figure.

posterior jaw discrepancy, and usually the greater the difficulty in correcting the malocclusion. Angles less than 2 degrees and readings less than zero (eg, -1 degree, -2 degrees, -3 degrees) indicate that the mandible is located ahead of the maxilla, suggesting a Class III skeletal relationship.

Occlusal plane

The occlusal plane is drawn through the region of the overlapping cusps of the first premolars and first molars.

A cephalometric survey of an orthodontic problem would be incomplete without an appraisal of the location of the teeth in occlusion to the face and the skull. Therefore, the angle of the occlusal plane to SN is measured. The mean reading for normal occlusions is 14 degrees (Fig 7-6).

Mandibular plane

The mandibular plane is drawn between gonion and gnathion. The mandibular plane angle is formed by relating it to the anterior cranial base (SN). The mean reading for this angle is 32 degrees (see Fig 7-6). Excessively high or low mandibular plane angles suggest unfavorable growth patterns in individuals. Such patterns may affect the outcome of treatment, and it is wise to anticipate such problems if they occur.

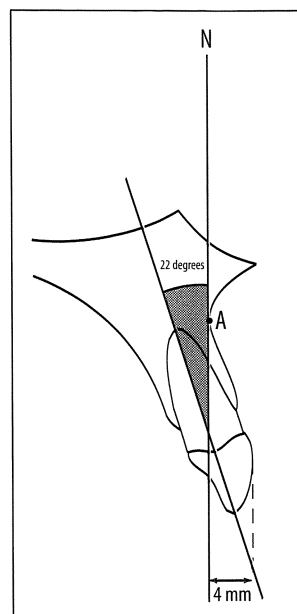


Fig 7-7 Maxillary incisor to NA line; 22 degrees and 4 mm is "ideal."

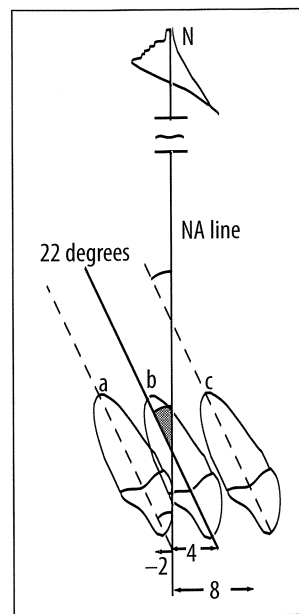


Fig 7-8 Incisor angled at 22 degrees but (a) retropositioned (-2 mm); (b) "ideally" positioned (4 mm); and (c) positioned too far forward (8 mm).

Dental analysis

The dental analysis usually confirms the clinical observations already made. However, there are numerous instances in which the radiographic picture differs markedly from the clinical concept of the location of the incisors.

Maxillary incisor position

The relative location and axial inclination of the maxillary incisors are determined by relating the teeth to the line from nasion to point A (NA). The maxillary incisor to NA reading in degrees indicates the relative angular relationship of the maxillary incisors, whereas the maxillary central incisor to NA reading in millimeters provides information on the forward or backward positioning of the incisors relative to the NA line (Fig 7-7).

Using this method, the maxillary central incisors should relate to the NA line in such a way that the most anteriorly placed point of its crown is 4 mm in front of the NA line and its axial inclination bears a 22-degree angle to the line. The use of linear and angular parameters in orienting the incisors provides information that relates to the location of the tooth anteroposteriorly to the NA line and to its angulation as well.

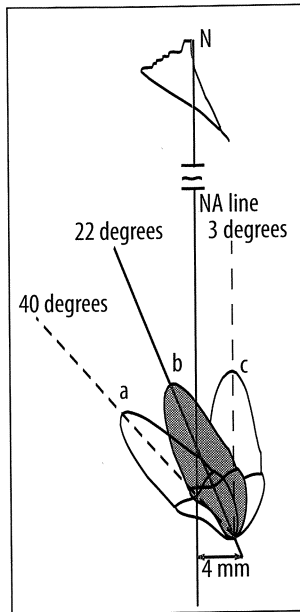


Fig 7-9 Illustration to show inadequacy of relating incisor tip to millimeter reading only. All three teeth are 4 mm from the NA line but angled differently (ie, 40 degrees, 22 degrees, and 3 degrees).

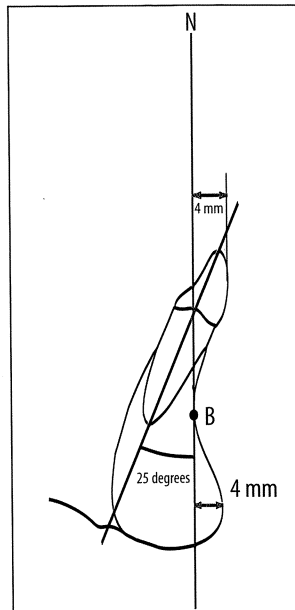


Fig 7-10 Relationship of mandibular incisor to NB line, 4 mm and 25 degrees.

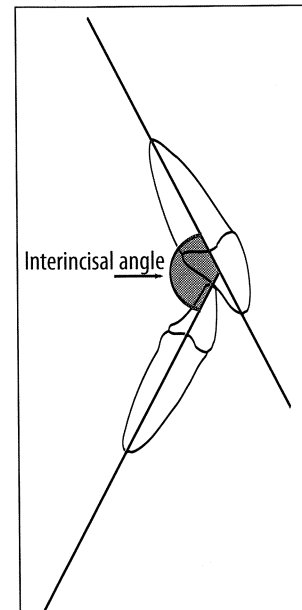


Fig 7-11 Interincisal angle.

The incisor angle alone does not convey adequate information relative to the anteroposterior position of this tooth in the facial complex. For example, the maxillary incisor may be angled at 22 degrees and ideally positioned anteroposteriorly (ie, 4 mm from the NA line) (Fig 7-8).

It is also possible for this tooth to be angled at 22 degrees to the NA line and to be positioned too far forward or backward in the facial skeleton. To precisely determine the relative anteroposterior position of the incisors, it is necessary to measure the distance of the most labial surface of the incisor to the NA line. In Fig 7-8, the angle of 22 degrees is the same in all three incisors outlined, but the relative anteroposterior relationship is correct in only the middle tooth in the tracing (see Fig 7-8, b). The remaining two incisors are either too far forward or backward (ie, the millimeter readings are either less or greater than 4 mm).

Similarly, a millimeter reading of only the maxillary incisor to the NA line is inadequate. An angular reading is necessary to indicate the degree of inclination of this tooth. It is not difficult to visualize a tooth, the labial surface (usually near the crown tip) of which is 4 mm from the NA line, as being either angled too vertically or inclined too labially (Fig 7-9).

Mandibular incisor position

The relative anteroposterior location and angulation of the mandibular incisors is determined by relating the teeth to the line from nasion to point B (NB). The mandibular incisor to NB measurement in millimeters shows the forward or backward positioning of these teeth relative to the NB line. The mandibular central incisor to NB reading in degrees indicates the relative axial inclination of these teeth. The most labial portion of the crown of the mandibular incisor should be located 4 mm ahead of the NB line, whereas the axial inclination of the tooth to this line should be 25 degrees (Fig 7-10). Ascertaining both the location and the angulation of the mandibular incisors is as important as in the case of the maxillary incisors.

Interincisal angle

The interincisal angulation relates the relative position of the maxillary incisor to that of the mandibular incisor. If the angle is more acute or less than the mean of 130 degrees (Fig 7-11), the maxillary or mandibular teeth (or both) often require uprighting. Conversely, if the angle is greater than 130 degrees or more obtuse, the maxillary or mandibular incisors (or both) often require advancing anteriorly or correcting of the axial inclination. The teeth causing the dis-

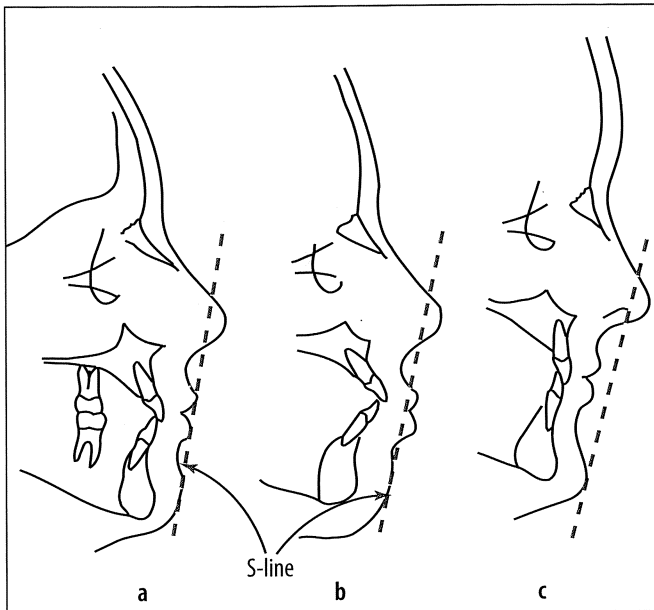


Fig 7-12 Steiner's S-line. (a) Lips in balance at rest; (b) lips too protrusive; (c) lips or lower facial profile too recessive.

crepancies in the readings can be determined by noting the relative angular positions of the maxillary teeth to NA or the mandibular teeth to NB.

Lower incisor–chin relationship

Since the chin contributes generously to the facial outline, this area must be evaluated. The degree of prominence of the chin contributes to the determination of the placement of the teeth in the arch. Ideally, according to Holdaway,⁵ the distance between the labial surface of the lower incisor to the NB line should be equal (ie, 4 mm) (see Fig 7-10). A 2-mm discrepancy between these measurements is acceptable; 3 mm is less desirable, but tolerable. If the difference between these dimensions exceeds 4 mm, however, corrective measures are generally indicated.

Soft tissue analysis

The soft tissue analysis is basically a graphic record of the visual observations made in the clinical examination of the patient. The soft tissue analysis includes an appraisal of the adaptation of soft tissue to the bony profile with consideration to the size, shape, and posture of the lips as seen on the lateral headfilm. The thickness of the soft tissue over the

symphysis mentalis and the nasal structure as it relates to the lower face is also analyzed.

Steiner, Ricketts, Holdaway, and Merrifield developed criteria and lines of reference for facial profile balance. Although there is no uniform concept of what constitutes an ideal profile, Steiner's S-line of reference for determining soft tissue facial balance is widely used in orthodontics today. The lips in well-balanced faces, according to Steiner, should touch a line extending from the soft tissue contour of the chin to the middle of an S formed by the lower border of the nose. This line is referred to as the *S-line* (Fig 7-12, a).

Lips located beyond this line tend to be protrusive (Fig 7-12, b), in which case the teeth and/or the jaws usually require orthodontic treatment to reduce the procumbency. If the lips are positioned behind this line, however, the patient's profile is generally interpreted as "concave" (Fig 7-12, c). Orthodontic correction usually entails advancing the teeth in the dental arches to build up the lips to approximate the S-line.

Interpretation of Cephalometric Headfilms

Figure 7-13 represents a tracing of a headfilm of a young white patient with a malocclusion. The various reference planes and lines were drawn, and the measurement for each was recorded in Table 7-1. Reference measurements for normal occlusion are listed in the middle column of the table.

The SNA reading of 88 degrees (normal mean, 82 degrees) suggests a protrusive maxilla. The SNB reading of 78 degrees (normal mean, 80 degrees) suggests a mildly recessive mandible. The ANB (difference between SNA and SNB) reading of 10 degrees (normal mean, 2 degrees) indicates a severe anteroposterior skeletal jaw dysplasia, which is attributed mainly to a protrusive maxilla.

The position of the maxillary incisors relative to the skeletal pattern (4 mm and 20 degrees) is good and does not need to be altered. The mandibular incisors, however, are severely tipped forward (ie, 12 mm and 45 degrees). Ideally, these should be uprighted to a position in which the reading would approximate those of "average" normal incisors (ie, 4 mm and 25 degrees). The interincisal angle is acute (104 degrees) due mainly to the severely protruded mandibular incisors.

The measurement of pogonion to NB is 0 mm. Relating this dimension to the mandibular incisors, which are tipped forward 12 mm, suggests that the incisors are in poor balance (1:12 mm ratio) to the skeletal pattern. To establish a 1:1 ratio, the mandibular incisors would have to be repositioned. This ratio would be easier to achieve if the patient had a chin

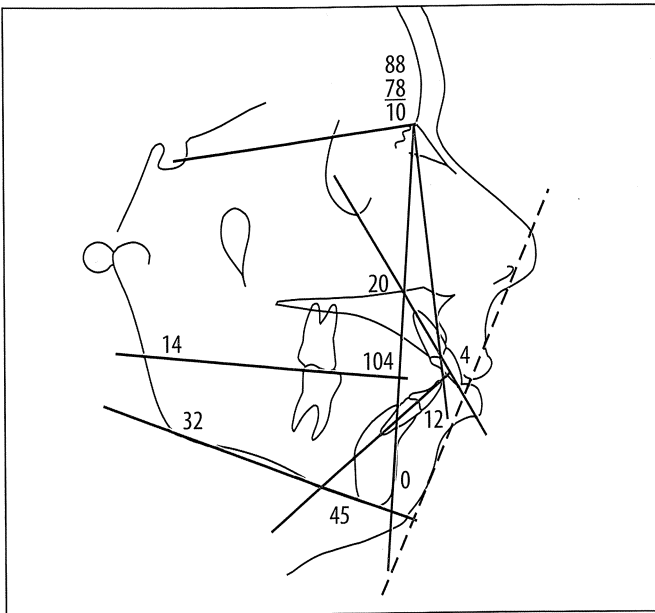


Fig 7-13 Tracing of patient with malocclusion.

Table 7-1 Steiner Analysis (Simplified) of Young White Patient with Malocclusion

Parameter	Reference measurements	Patient's measurements
SNA (angle)	82 degrees	88 degrees
SNB (angle)	80 degrees	78 degrees
ANB (angle)	2 degrees	10 degrees
1 to NA (mm)	4 mm	4 mm
1 to NA (angle)	22 degrees	20 degrees
T to NB (mm)	4 mm	12 mm
T to NB (angle)	25 degrees	45 degrees
1 to T (angle)	131 degrees	104 degrees
Pog to NB (mm)	Not established	0 mm
Pog and T to NB (difference)	Not established	12 mm
OP to SN (angle)	14 degrees	14 degrees
GoGn to SN (angle)	32 degrees	32 degrees

with a large symphysis mentalis. The measurements of the occlusal and mandibular planes in this tracing, however, approximate normal values and, as such, are satisfactory.

The S-line in this tracing shows that the lips are too protrusive. Bodily retraction of the maxillary incisors and lingual tipping of the mandibular incisors would reduce the protrusiveness of these lips. Forward growth of the mandible would also help improve the profile.

Cephalometric analyses of skeletal, dental, and soft tissues are merely aids in determining the diagnosis. For an accurate interpretation, the various readings must not be assessed independently. It cannot be too strongly emphasized, however, that to interpret the data, all measurements must be correlated with other clinical and diagnostic criteria before arriving at the diagnosis and treatment plan.

Acceptable Compromises

Cephalometry must not be regarded as a numbers game in which the measured parameters of the tracing must approximate those of normal occlusions or the dentofacial skeletal

pattern will be regarded as being imbalanced. In attempting to simplify the presentation of their ideas, clinicians have developed a set of figures as a mean (these figures are to be varied by judgment as is indicated for the individual). No proponent of any analysis has ever suggested that every individual should conform to one set of measurements. Variation in biology is the rule rather than the exception. Normal is never a point; it is a range.

Downs presented a mean for such measurements and wisely provided positive and negative limits (a range) within which measurements of individuals can vary and still be within the range of normalcy. Wylie,⁶ however, made it clear that variations within these limits must occur in the right combinations if the individual is to appear normal. Judgment is still necessary to decide if the combinations of these variations are desirable.

The orthodontist has considerable control in the correction or repositioning of malaligned teeth. The cephalometric headfilm provides information regarding the degree or extent of tooth movement necessary to accomplish or recover harmony in a malocclusion. Knowledge of variations of the relationship of the dentition to the skeletal pattern in individuals with excellent occlusions is thus an aid to locating areas of disharmony in malocclusion.

Skeletal patterns, however, are relationships over which we have little control in orthodontic treatment. During growth, there are varying degrees of downward and forward growth of the face relative to the cranial base. The degree to which this growth can be harnessed during orthodontic therapy is still a subject of much debate. It is sufficient to say that the manner in which the face grows during and after treatment has significant bearing on the prognosis of a patient. Many of the difficulties experienced in treatment can be attributed directly to the extent of the excessive disharmony of the skeletal pattern. It is better to recognize skeletal disharmony by means of a cephalometric radiograph before treatment and alert the patient than to be embarrassed by the discovery of these difficulties months after treatment has begun.

Cephalometric radiographs taken serially during the course of treatment often provide information on the extent to which treatment objectives are being achieved in the restoration of harmony and balance to the component parts of the face. Such radiographs assist in clarifying the possibilities and limitations of the advocated treatment procedure.

Steiner clearly recognized that cephalometric standards are merely gauges by which to determine more favorable compromises as a treatment goal. He developed a chart that reflects a number of average measurements of normal dentofacial relationships. Not all anteroposterior skeletal discrepancies can be orthodontically corrected to a so-called ideal jaw relationship. For example, the likelihood of reducing an ANB discrepancy of 10 degrees to that of an average normal of 2 degrees by means of orthodontic correction, even if aided by growth, is almost nil. However, with treatment, it may be possible to reduce the antero-

posterior discrepancy (ANB angle) from 10 degrees to 6 degrees or maybe even 5 degrees. This is not to infer that because of the jaw relationship (ANB of 5 or 6 degrees) the teeth in the respective arches cannot be made to occlude satisfactorily. On the contrary, in such instances, if the maxillary incisors were inclined a little more lingually and the mandibular incisors a little more labially than the "ideal" relationship (ie, 4 mm and 22 degrees for maxillary incisors and 4 mm and 25 degrees for mandibular incisors), a well-balanced and harmonious occlusion could be achieved. Steiner thus developed a series of acceptable compromise measurements for which patients could be treated and yet possess excellent dentofacial harmony and balance. (For details of the application of these compromise measurements in the planning and assessing of orthodontic cases, the reader is referred to the original publication by Steiner.³)

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